Q2. [14 pts] Principal Components Analysis

Consider the following design matrix, representing four sample points $X_i \in \mathbb{R}^2$.

$$X = \begin{bmatrix} 4 & 1 \\ 2 & 3 \\ 5 & 4 \\ 1 & 0 \end{bmatrix}.$$

We want to represent the data in only one dimension, so we turn to principal components analysis (PCA).

(1) [5 pts] Compute the **unit-length principal component directions** of X, and **state which one the PCA algorithm would choose** if you request just one principal component. Please provide an exact answer, without approximation. (You will need to use the square root symbol.) **Show your work!**

We center X, yielding

$$\dot{X} = \begin{bmatrix} 1 & -1 \\ -1 & 1 \\ 2 & 2 \\ -2 & -2 \end{bmatrix}.$$

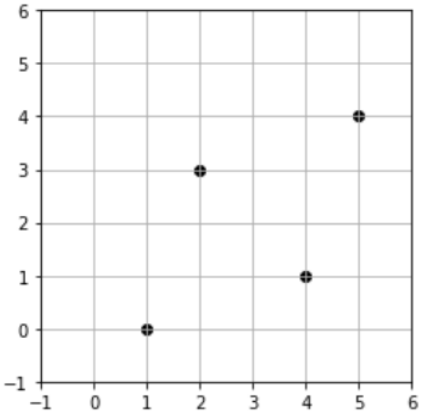
Then $\dot{X}^\top \dot{X} = \begin{bmatrix} 10 & 6 \\ 6 & 10 \end{bmatrix}$. (Divide by 4 if you want the sample covariance matrix. But we don't care about the magnitude.)

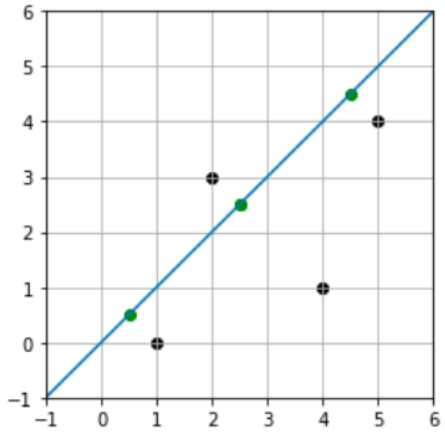
Its eigenvectors are $[1/\sqrt{2} \quad 1/\sqrt{2}]^\top$ with eigenvalue 16 and $[1/\sqrt{2} \quad -1/\sqrt{2}]^\top$ with eigenvalue 4. The former eigenvector is chosen.

(Negated versions of these vectors also get full points.)

(2) [5 pts] The plot below depicts the sample points from X. We want a one-dimensional representation of the data, so **draw the principal component direction (as a line) and the projections of all four sample points onto the principal direction**.

Label each projected point with its principal coordinate value (where the origin's principal coordinate is zero). Give the principal coordinate values exactly.



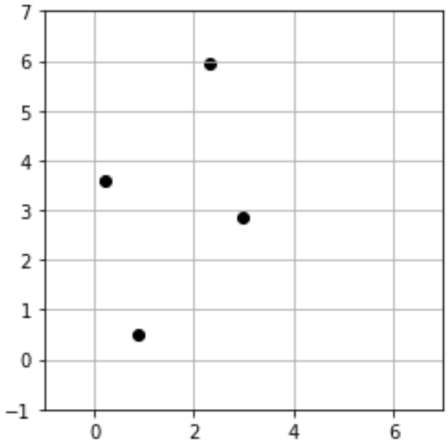


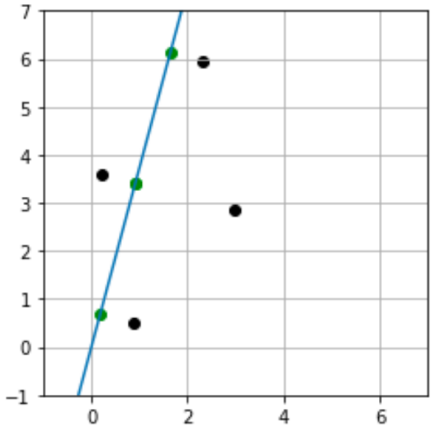
The principal coordinates are $\frac{1}{\sqrt{2}}$, $\frac{5}{\sqrt{2}}$, $\frac{5}{\sqrt{2}}$, and $\frac{9}{\sqrt{2}}$. (Alternatively, all of these could be negative, but they all have to have the same sign.)

(3) [4 pts] The plot below depicts the sample points from X rotated 30 degrees counterclockwise about the origin.

As in part (b), **identify the principal component direction that the PCA algorithm would choose and draw it (as a line) on the plot**. Also **draw the projections of the rotated points** onto the principal direction.

Label each projected point with the exact value of its principal coordinate.





The line passes through the origin and is parallel to the two sample points that are farthest apart, so it's easy to draw. Rotation has not changed the principal coordinates: $\frac{1}{\sqrt{2}}$, $\frac{5}{\sqrt{2}}$, $\frac{5}{\sqrt{2}}$, and $\frac{9}{\sqrt{2}}$. (Again, these could all be negative.)

Q3. [14 pts] A Decision Tree

In this question we investigate whether students will pass or fail CS 189 based on whether or not they studied, cheated, and slept well before the exam. You are given the following data for five students. There are three features, "Studied," "Slept," and "Cheated." The column "Result" shows the label we want to predict.

	Studied	**Slept**	**Cheated**	**Result**
Student 1	Yes	No	No	Passed
Student 2	Yes	No	Yes	Failed
Student 3	No	Yes	No	Failed
Student 4	Yes	Yes	Yes	Failed
Student 5	Yes	Yes	No	Passed

(1) [4 pts] What is the **entropy** $H(\text{Result})$ at the root node? (There is no need to compute the exact number; you may write it as an arithmetic expression.)

$$H(\text{Result}) = -\left(\frac{2}{5}\log_2\frac{2}{5} + \frac{3}{5}\log_2\frac{3}{5}\right).$$

(2) [5 pts] **Draw the decision tree** where every split maximizes the information gain. (An actual drawing, please; a written description does not suffice.) Do not perform a split on a pure leaf or if the split will produce an empty child; otherwise, split. **Explain (with numbers)** why you chose the splits you chose.

A tree that first splits on "Cheated" and then "Studied."

(3) [2 pts] Did the tree you built implicitly perform feature subset selection? **Explain.**

Yes, because it does not use the feature "Slept."

(4) [3 pts] Suppose you have a sample of n students for some large n, with the same three features. Assuming that we use a reasonably efficient algorithm to build the tree (as discussed in class), what is the **worst-case running time to build** the decision tree? (Write your answer in the simplest asymptotic form possible.) **Why?**

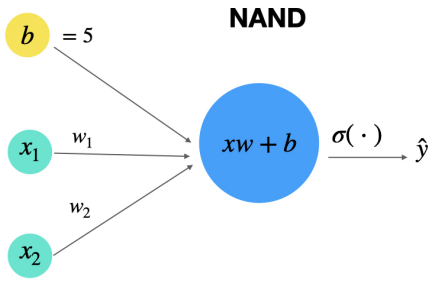
We have 3 binary features, so the tree's depth cannot exceed 3 and each sample point participates in at most four treenodes. Hence, it cannot take more than $\Theta(n)$ time to build the tree.

Q6. [10 pts] A Miscellany

(1) [4 pts] Consider a single unit in a neural network that receives two binary inputs $x_1, x_2 \in \{0, 1\}^2$ and computes a linear combination followed by a threshold activation function, namely,

$$\sigma(z) = \begin{cases} 1, & z \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

The unit is illustrated below. We have chosen a bias term of $b = 5$. **Provide values for the two weights w_1 and w_2** that allow you to compute the NAND function (which is 0 if and only if both inputs are 1).



$w_1 = -3$, $w_2 = -3$ will work.

(2) [6 pts] We are drawing sample points from a distribution with the probability density function (PDF) $f(x) = \frac{1}{2}\, e^{-|x-\mu|}$, but we do not know the mean $\mu \in \mathbb{R}$. We decide to estimate μ with maximum likelihood estimation (MLE). Unfortunately, we have only two sample points $X_1, X_2 \in \mathbb{R}$.

Derive the likelihood and the log-likelihood for this problem. Then **show that every value of μ between X_1 and X_2 is a maximum likelihood estimate**.

The likelihood is

$$\mathcal{L}(\mu; X_1, X_2) = \frac{1}{4} e^{-|X_1-\mu|} e^{-|X_2-\mu|},$$

and the log-likelihood is

$$\ell(\mu; X_1, X_2) = -|X_1 - \mu| - |X_2 - \mu| - \ln 4.$$

For any μ between X_1 and X_2 (inclusive), the log-likelihood is $-|X_1 - X_2| - \ln 4$. For any μ outside that range, it is lesser. (For example if μ is less than $\min\{X_1, X_2\}$, then the log-likelihood is $-|X_1 - X_2| - 2|\mu - \min\{X_1, X_2\}| - \ln 4$).